

VR-Algebra — v1.0.1 (revision note)

Synchronisation with VR v1.1.1 (one-line correction of the parent description and DOI)

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Scope. This is the revision delta for VR-Algebra v1.0.1. *No structure, definition, theorem, finding, or axiom-profile of VR-Algebra is altered.* The single change is editorial: the recall of the parent work VR (and its citation) is brought up to date with VR v1.1.1.

The correction

The preliminaries described VR as

“VR. A Formal System (Reznik, 2026; Zenodo DOI 10.5281/zenodo.20212092) — a minimal axiomatisation of arithmetic on **three primitives** $\{\emptyset, \rightarrow, t\}$ and **four axioms**, equivalent to Peano arithmetic.”

This is superseded on two counts. (i) **The primitive count and the axioms.** In VR v1.1.1 the propositional implication $\rightarrow$ (and the principles A1, A2 and the two-valued logical register) are **removed**: $\rightarrow$ generated only a finite truth-function algebra used by no VR theorem — arithmetic, not logic — and VR generates no logic of its own. (ii) **The DOI** cited (20212092) is the original v1.0; the current version is v1.1.1. The corrected recall reads

“VR. A Formal System (Reznik, 2026; Zenodo DOI 10.5281/zenodo.20688164, v1.1.1; concept 10.5281/zenodo.20212091) — a minimal *axiom-free* reconstruction of arithmetic on **two primitives** $\{\emptyset, t\}$, with no axioms of its own, equivalent to first-order Peano arithmetic.”

Why VR-Algebra is otherwise unaffected

VR-Algebra (the operational tracking discipline applied to algebraic structures) does not depend on the removed layer: it references VR only as the foundational arithmetic, through the operational reading and the nullary base $\emptyset$, unchanged in v1.1.1. The Lean module (`VRCycle.Algebra`) contains no reference to the removed objects (`VRBool`, `impl`, A1, A2), and the build and axiom profiles are unaffected.

Prepared with the assistance of Claude (Anthropic, Opus 4.8) in the cycle’s Variant A architecture; all mathematical content and positions are due to the author, Vitaly Reznik.